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Seattle vs. Vancouver: A Statistical Comparison of Precipitation by Season and Month

Introduction

Seattle has a reputation for being rainy and foggy, while Vancouver is often perceived as having milder weather. However, the two cities are geographically close and share similar climates, raising the question of whether Seattle's reputation for rain is justified.

This project investigates whether Seattle receives more precipitation than Vancouver and whether either city has an undeserved reputation for being rainy. We define "rains more" in two ways: the frequency of rainy days and the amount of precipitation. Using historical weather data, we compare the two cities across seasons and months to examine whether differences in precipitation depend on the time of year.

Data

The data for this project were obtained from the National Oceanic and Atmospheric Administration (NOAA) GHCN Daily database. The dataset contains daily precipitation records measured in inches for Seattle (station: US1WAKG0225) and Vancouver (station: CA001108395). The study period covers 1,826 days for each of the two cities from January 1, 2018, to December 31, 2022. Of these, 3 (0.1%) had no recorded precipitation value. These missing values were imputed using the mean precipitation of the same calendar day across the other years.

Analysis method

Simply comparing the mean precipitation of Seattle and Vancouver cannot show whether the difference is real or merely due to chance, such as a single outlier day. I therefore conducted two hypothesis tests to determine whether there is a statistically significant difference in precipitation between the two cities.

Test 1: Amount of precipitation. To determine whether Seattle receives less precipitation than Vancouver, I conducted a one-sided two-sample t-test, which compares the means of two groups. Daily precipitation is right-skewed because many days have zero precipitation. However, the sample size is large enough for the Central Limit Theorem to apply, so the t-test remains appropriate. I chose a one-sided test because the monthly box plots showed that Seattle had lower precipitation than Vancouver in more months.

Test 2: Frequency of rainy days. To examine how often it rains, I created a Boolean column that is True when daily precipitation is greater than 0. Using this data, I computed the proportion of rainy days in each city and conducted a one-sided two-sample z-test for proportions. While the t-test compares how much it rains, the z-test compares how often it rains. I again chose a one-sided test in the opposite direction because Seattle's overall proportion of rainy days was slightly higher than Vancouver's.

Result

Figure 1 and 2 show the mean daily precipitation of Seattle and Vancouver by season and month. Stars mark the periods in which Seattle's mean precipitation was significantly lower than Vancouver's, based on one-sided two-sample t-tests at α = 0.05. Only Fall and November were significant (p = 0.004 and p = 0.025, respectively). In the remaining seasons and months, the difference was not statistically significant.

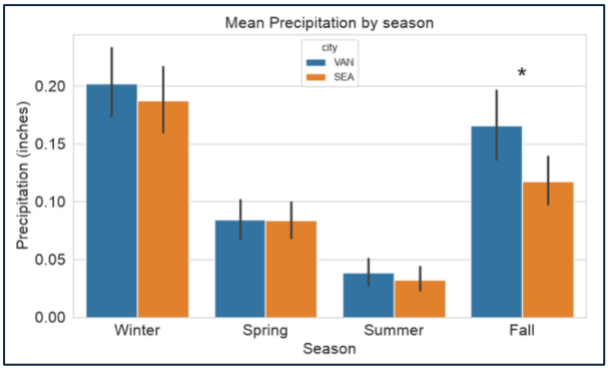


Figure 1

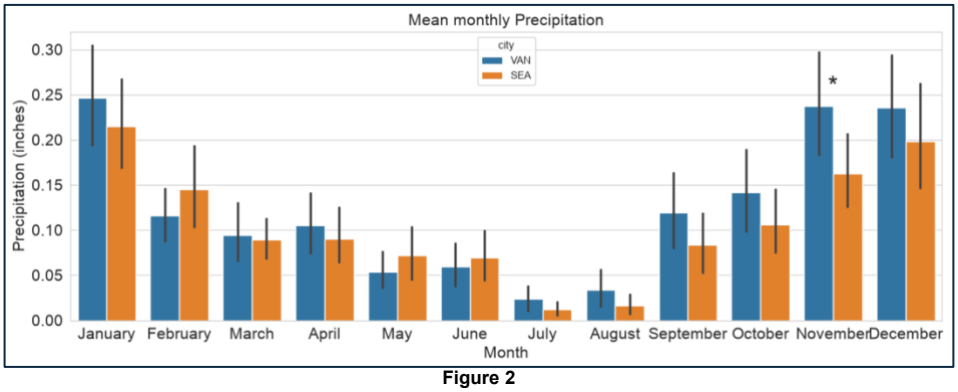


Figure 2

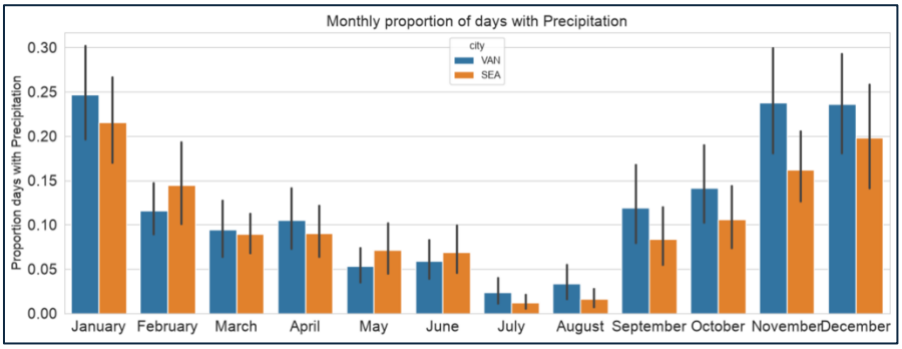


Figure 3

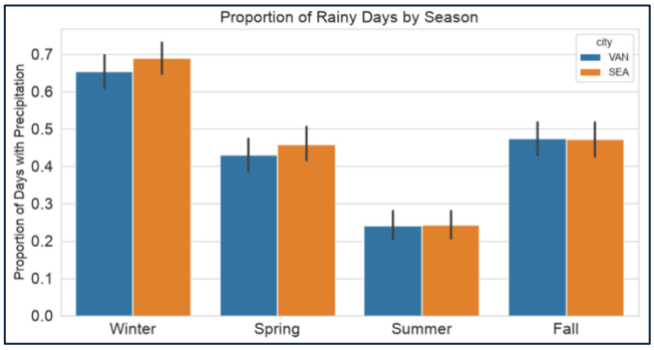


Figure 4

In contrast, the z-tests for proportions found no significant difference in the frequency of rainy days between the two cities either by month or by season at α = 0.05. Although the monthly bar plots (Figure 3) show some visible differences in the proportion of rainy days, the z-test on the overall proportion did not find a statistically significant difference. This is likely because the overall difference is small relative to its standard error. In winter and spring, Seattle had a slightly higher proportion of rainy days than Vancouver, but the difference was not statistically significant (p = 0.128 and p=0.194, respectively).

Conclusion

Seattle had significantly lower mean precipitation than Vancouver in fall and November, while the proportion of rainy days did not differ significantly between the two cities.

Future research could define a threshold for heavy precipitation and compare the proportion of days exceeding it in each city. This would help determine whether Seattle's reputation for being rainy is justified in terms of both frequency and intensity.